

PBL2_L4.R

Harshal

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```
# Lab 4 Advanced missing data handling
```

```
required_packages <- c("naniar", "skimr", "ggplot2")
for (pkg in required_packages) {
  if (!require(pkg, character.only = TRUE)) install.packages(pkg, dependencies = TRUE)
  library(pkg, character.only = TRUE)
}
```

```
## Loading required package: naniar
```

```
## Loading required package: skimr
```

```
##
```

```
## Attaching package: 'skimr'
```

```
## The following object is masked from 'package:naniar':
```

```
##
```

```
##     n_complete
```

```
## Loading required package: ggplot2
```

```
url <- "https://archive.ics.uci.edu/ml/machine-learning-databases/adult/adult.data"
adult_cols <- c("age", "workclass", "fnlwgt", "education", "education_num",
               "marital_status", "occupation", "relationship", "race", "sex",
               "capital_gain", "capital_loss", "hours_per_week", "native_country", "income")
```

```
data <- read.csv(url, header = FALSE, col.names = adult_cols, strip.white = TRUE, stringsAsFactors = FALSE)
```

```
set.seed(47)
```

```
n <- nrow(data)
```

```
data$age[sample(1:n, 1500)] <- 999
```

```
data$age[sample(1:n, 2000)] <- NA
```

```
data$workclass[sample(1:n, 3500)] <- ""
```

```
data$hours_per_week[sample(1:n, 800)] <- NaN
```

```
cat("Count of NA in age:", sum(is.na(data$age)), "\n")
```

```
## Count of NA in age: 2000
```

```

cat("Count of NaN in hours_per_week:", sum(is.nan(data$hours_per_week)), "\n")

## Count of NaN in hours_per_week: 800

cat("Count of blank strings (\"\") in workclass:", sum(data$workclass == ""), "\n")

## Count of blank strings (\"") in workclass: 3500

cat("Count of impossible age values (age == 999):", sum(data$age == 999, na.rm = TRUE), "\n")

## Count of impossible age values (age == 999): 1408

dummy_obj <- NULL
cat("is.null(dummy_obj):", is.null(dummy_obj), "\n")

## is.null(dummy_obj): TRUE

cat("is.null(data$age[1]):", is.null(data$age[1]), "\n")

## is.null(data$age[1]): FALSE

cat("\nVariable-wise missing summary:\n")

##
## Variable-wise missing summary:

print(miss_var_summary(data))

## # A tibble: 15 x 3
##   variable      n_miss pct_miss
##   <chr>         <int>   <num>
## 1 age           2000     6.14
## 2 hours_per_week 800     2.46
## 3 workclass       0      0
## 4 fnlwgt         0      0
## 5 education       0      0
## 6 education_num   0      0
## 7 marital_status  0      0
## 8 occupation      0      0
## 9 relationship    0      0
## 10 race           0      0
## 11 sex            0      0
## 12 capital_gain   0      0
## 13 capital_loss    0      0
## 14 native_country 0      0
## 15 income         0      0

```

```

impute_median <- function(x) {
  if (!is.numeric(x)) stop("Input must be a numeric vector.")
  valid_median <- median(x[!is.na(x) & !is.nan(x)], na.rm = TRUE)
  x[is.na(x) | is.nan(x)] <- valid_median
  return(x)
}

cleaned_data <- data
cleaned_data$age[cleaned_data$age == 999] <- NA
cleaned_data$workclass[cleaned_data$workclass == ""] <- "Unknown"
cleaned_data$occupation[cleaned_data$occupation == "?"] <- "Unknown"
cleaned_data <- cleaned_data[!is.nan(cleaned_data$hours_per_week), ]
cleaned_data$age <- impute_median(cleaned_data$age)
cleaned_data$hours_per_week <- impute_median(cleaned_data$hours_per_week)
complete_rows <- sum(complete.cases(cleaned_data))
cat("\nNumber of completely intact rows:", complete_rows, "out of", nrow(cleaned_data), "\n")

```

```

##
## Number of completely intact rows: 31761 out of 31761

```

```

cat("\nVariable-wise missing summary (After Cleaning):\n")

```

```

##
## Variable-wise missing summary (After Cleaning):

```

```

print(miss_var_summary(cleaned_data))

```

```

## # A tibble: 15 x 3
##   variable      n_miss pct_miss
##   <chr>         <int>    <num>
## 1 age           0         0
## 2 workclass     0         0
## 3 fnlwgt       0         0
## 4 education     0         0
## 5 education_num 0         0
## 6 marital_status 0         0
## 7 occupation    0         0
## 8 relationship  0         0
## 9 race         0         0
## 10 sex         0         0
## 11 capital_gain 0         0
## 12 capital_loss 0         0
## 13 hours_per_week 0         0
## 14 native_country 0         0
## 15 income      0         0

```

```

# task 5
skim_summary <- skim_without_charts(cleaned_data)
print(skim_summary)

```

```

## -- Data Summary -----

```

```

##                               Values
## Name                         cleaned_data
## Number of rows                31761
## Number of columns             15
## -----
## Column type frequency:
##   character                    9
##   numeric                      6
## -----
## Group variables                None
##
## -- Variable type: character -----
##   skim_variable  n_missing complete_rate min max empty n_unique whitespace
## 1 workclass      0           1  1  16      0      10           0
## 2 education      0           1  3  12      0      16           0
## 3 marital_status 0           1  7  21      0       7           0
## 4 occupation     0           1  5  17      0      15           0
## 5 relationship   0           1  4  14      0       6           0
## 6 race           0           1  5  18      0       5           0
## 7 sex            0           1  4   6      0       2           0
## 8 native_country 0           1  1  26      0      41           0
## 9 income         0           1  4   5      0       2           0
##
## -- Variable type: numeric -----
##   skim_variable  n_missing complete_rate   mean      sd    p0    p25    p50
## 1 age            0           1    38.5    13.0    17    29    37
## 2 fnlwgt         0           1 189680. 105533. 12285 117746 178356
## 3 education_num  0           1    10.1     2.57     1     9    10
## 4 capital_gain   0           1   1085.    7440.     0     0     0
## 5 capital_loss   0           1    87.4    403.     0     0     0
## 6 hours_per_week 0           1    40.4    12.3     1    40    40
##   p75    p100
## 1    46     90
## 2 237141 1484705
## 3    12     16
## 4     0  99999
## 5     0   4356
## 6    45     99

cat("Impossible age (999) remaining:", sum(cleaned_data$age == 999, na.rm = TRUE), "\n")

## Impossible age (999) remaining: 0

cat("Untreated NAs in age:", sum(is.na(cleaned_data$age)), "\n")

## Untreated NAs in age: 0

cat("Blank strings remaining in workclass:", sum(cleaned_data$workclass == ""), "\n")

## Blank strings remaining in workclass: 0

```

```
write.csv(cleaned_data, "cleaned_data.csv", row.names = FALSE)
cat("\nSaved output file as 'cleaned_data.csv'\n")
```

```
##
## Saved output file as 'cleaned_data.csv'
```

```
library(naniar)
library(gridExtra)

p_before <- vis_miss(data) + ggtitle("Data Missingness Matrix (Before)")
p_after  <- vis_miss(cleaned_data) + ggtitle("Data Missingness Matrix (After)")

png("missingness_matrix_comparison.png", width = 1000, height = 500)
grid.arrange(p_before, p_after, ncol = 2)
dev.off()
```

```
## pdf
## 2
```